DEPARTMENT OF PHYSICS

Indian Institute of Technology (BHU) Varanasi

Spin Wave Based Band-Pass and Band-Stop Filters with Structurally Modulated Waveguides

EP-291 Project Report

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Table of Contents

Abstract 2
1. Introduction 3
1.1 Reference Paper 4
2. Theoretical Background 4
2.1 Spin Waves and Magnons 4
2.2 Landau–Lifshitz–Gilbert (LLG) Equation 4
2.3 Magnonic Crystals and Band Engineering 5
2.4 Filter Topologies 5
3. Spin Waves in Next-Generation Technology 5
4. Simulation Methodology 6
4.1 Simulation Software: MuMax3 6
4.2 Material Parameters 6
4.3 Waveguide Geometry 6
4.4 Excitation and Transmission Calculation 7
4.5 Code Scripts 7
5. Results and Analysis — Replication of Feng et al. 8
5.1 Transmission Spectra 8
5.2 Temporal Snapshots 9
5.3 Spin-Wave Precession Animation 10
6. Discussion 10
7. Future Prospects 11
8. Conclusions 11
References 12
Acknowledgements 12

Abstract

This report presents the replication and validation of results from the paper by Feng et al. (Chinese Academy of Sciences) on spin-wave based band-pass and band-stop filters utilizing structurally modulated magnonic waveguides. The primary objective of this exploratory project was to independently reproduce the key numerical findings of that work using micromagnetic simulation tools, thereby developing a working understanding of spin-wave physics and magnonic device design. Spin waves — collective oscillatory excitations of magnetic spins in ferromagnetic media — represent a promising platform for next-generation signal processing devices operating in the gigahertz frequency range. The study employs the Landau–Lifshitz–Gilbert (LLG) formalism to model magnetization dynamics. Transmission spectra confirming threshold frequencies at 21 GHz and 25 GHz are obtained, alongside temporal snapshots and precession animations at selected test frequencies of 44.9 GHz and 49.1 GHz, all in close agreement with the results reported by Feng et al. The findings demonstrate the feasibility of compact, low-power magnonic filter devices with applications in energy-efficient communication, neuromorphic computing, and hybrid CMOS–magnonic systems.

1. Introduction

The rapid miniaturization of electronic devices has driven intense research into beyond-CMOS technologies capable of overcoming the limitations of conventional charge-based logic and signal processing. Among the emerging paradigms, magnonics — the field exploiting spin waves (magnons) as information carriers — has attracted significant attention due to its potential for extremely low power operation, sub-micron device footprints, and inherent support for wave-based parallel computation.

This exploratory project (EP 291) is centered on the replication of key simulation results from the research paper by Feng et al. from China, which demonstrated spin-wave based band-pass and band-stop filter designs using structurally modulated magnonic waveguides. The primary goal was to reproduce the transmission spectra, temporal snapshots, and precession behavior reported in that paper through independent micromagnetic simulations, and in doing so, gain a thorough understanding of spin-wave physics and magnonic device modeling.

Spin waves are propagating disturbances of the magnetic spin alignment in ordered magnetic materials. These waves can be excited, guided, and manipulated within nano-scale waveguides, making them ideal for implementing filtering, logic, and modulation functions at microwave and millimeter-wave frequencies. Unlike photonic or acoustic analogues, spin-wave devices can be made reconfigurable via applied magnetic fields, adding a degree of programmability not easily achieved in solid-state electronics.

This project investigates the design of band-pass and band-stop filter configurations using structurally modulated magnonic waveguides, following the methodology of Feng et al. Structural modulation — achieved by periodically varying the waveguide geometry or material composition — introduces band gaps in the spin-

wave dispersion relation, enabling selective frequency transmission and rejection. The project culminates in replicated simulation results that confirm the targeted filter responses originally reported in the reference paper.

1.1 Reference Paper

The work in this project is based on the following paper-

Lai-He Feng, Mang-Yuan Ma, Zhi-Hua Liu, Kai-Le Xie, Fu-Sheng Ma

School of Physics and Technology, Nanjing Normal University, Nanjing 210046, China

"Magnonic band-pass and band-stop filters with structurally modulated waveguides"

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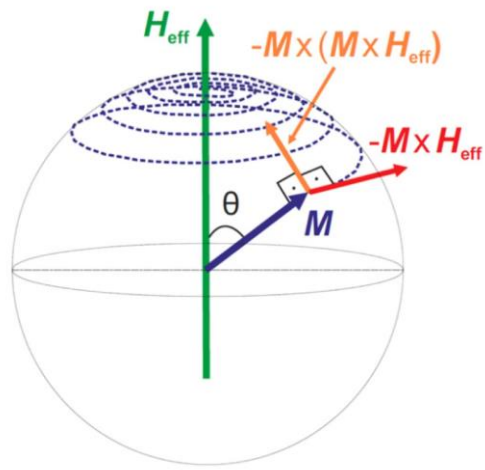
The paper presents MuMax3 micromagnetic simulations of CoFeB magnonic filter structures with perpendicular magnetization. The band-stop filter (uniformly magnetized modulator) exhibits a transmission threshold near 25 GHz, with test frequencies at 44.9 GHz (T = 9%, stop-band) and 49.1 GHz (T = 92%, pass-through). This project independently replicates the band-stop filter results, obtaining a threshold at 21 GHz.

2. Theoretical Background

2.1 Spin Waves and Magnons

Spin waves arise from the exchange and dipolar interactions between neighboring magnetic moments in a ferromagnetic lattice. When a magnetic system is disturbed from its equilibrium configuration, the perturbation propagates as a wave. The quantum of this excitation is the magnon, analogous to the phonon in lattice vibrations. Spin-wave wavelengths can be as small as tens of nanometers, enabling device dimensions far below those achievable with electromagnetic waves of equivalent frequency.



2.2 Landau–Lifshitz–Gilbert (LLG) Equation

The dynamics of the magnetization vector M in a ferromagnetic medium are governed by the Landau–Lifshitz–Gilbert (LLG) equation-

$$dM/dt = -\gamma\,(M \times H_E^{ff}) + (\alpha/M_s)(M \times dM/dt)$$

where the key parameters are defined as follows-

- γ (Gyromagnetic Ratio)- Proportionality constant relating angular momentum to magnetic moment (~1.76×10^{11} rad s^{-1} T^{-1} for electrons)
- α (Damping Constant)- Dimensionless Gilbert damping parameter governing energy dissipation
- M (Magnetisation)- Local magnetization vector of the material
- M_s (Saturation Magnetisation)- Maximum magnetization achievable in the material
- μ_o (Permeability of Free Space)- $4\pi \times 10^{-7}$ H/m
- H_E^{ff} (Effective Magnetic Field)- Includes exchange, dipolar, anisotropy, and applied field contributions

The first term on the right-hand side describes the precessional torque, causing M to precess around H_E^{ff}. The second term is the damping torque, driving M back toward the equilibrium direction. The balance between these terms determines both the frequency and the decay length of spin-wave modes.

2.3 Magnonic Crystals and Band Engineering

Magnonic crystals are periodic magnetic structures analogous to photonic crystals. By modulating the geometry or composition of a magnetic waveguide with a spatial period comparable to the spin-wave wavelength, one can open band gaps — frequency ranges in which spin-wave propagation is forbidden. These

band gaps underpin the filter functionality- frequencies within the gap are reflected or evanescent (band-stop behavior), while frequencies outside the gap propagate freely (band-pass behavior).

2.4 Filter Topologies

Two principal filter configurations are investigated-

- **Band-Pass Filter- Transmits a selected band of frequencies while attenuating others. Achieved by engineering a pass-band in the spin-wave dispersion, surrounded by band gaps on either side.**
- **Band-Stop (Notch) Filter- Rejects a specific frequency band and transmits all others. Realized by placing the target frequencies inside a magnonic band gap.**

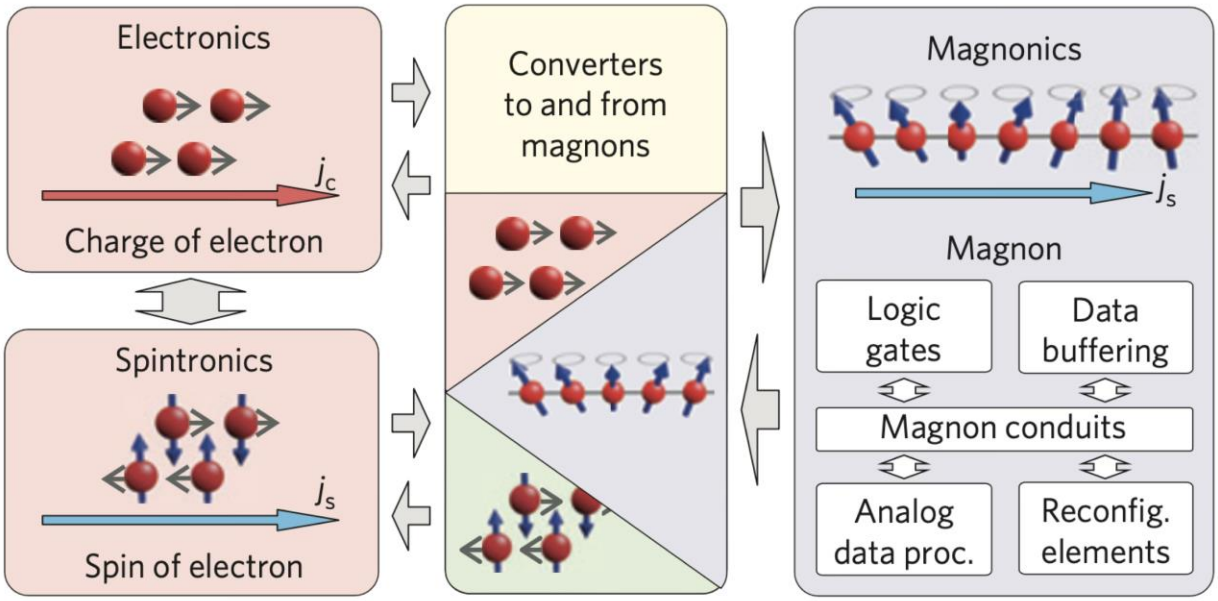
3. Spin Waves in Next-Generation Technology

Spin-wave based devices represent a convergence of condensed-matter physics and electrical engineering. Several proof-of-concept demonstrations have established their technological viability-

- Spin-Wave Transistor- Analogous to the electronic transistor, a spin-wave transistor modulates the amplitude or phase of a propagating spin wave through a gate field, enabling logical switching with sub-femtojoule energy dissipation.
- Magnonic XOR Gate- Demonstrated using magnonic crystal structures where interference between spin-wave beams implements Boolean logic. The XOR function has been realized by exploiting the phase sensitivity of spin-wave superposition.
- Spin-Wave Computing Flow- A complete computing paradigm has been proposed in which data is encoded in spin-wave amplitude or phase, processed through magnonic logic gates, and read out via inverse spin-Hall or magneto-optical detection.

The key advantages of spin-wave devices over CMOS alternatives include-

Advantage	Significance
Extremely Small Footprint	Spin-wave devices can be realized at the nanometer scale, enabling ultra-dense integration.
Low Power Operation	Wave-based computation avoids charge transport, drastically reducing Joule heating.
Parallel Computing	Multiple spin-wave modes can propagate simultaneously in a single waveguide, enabling wave-division multiplexing of data.
Reconfigurability	Applied magnetic fields can dynamically tune the dispersion relation, enabling programmable filter responses.



4. Simulation Methodology

4.1 Simulation Software- MuMax3

All micromagnetic simulations were performed using MuMax3, a GPU-accelerated open-source software package that solves the Landau–Lifshitz–Gilbert (LLG) equation using the finite difference method. MuMax3 is widely used in the magnonics community for its high computational efficiency and its ability to handle complex three-dimensional magnetic textures including skyrmions. The same software was used by Feng et al. in the reference paper, making it the natural choice for replication.

4.2 Material Parameters

Following Feng et al., the magnetic parameters used in the simulation were derived from CoFeB, a material with strong perpendicular magnetic anisotropy suitable for out-of-plane magnetized filter devices. The parameters are listed below-

Parameter	Symbol	Value

Saturation Magnetisation	Ms	1.6×10^6 A/m
Exchange Constant	A	16×10^{-12} J/m
Perpendicular Anisotropy	Ku	2×10^6 J/m^3
DMI Constant	D	3.0 mJ/m^2
Gilbert Damping	α	0.001
Cell Size	—	1 nm × 1 nm × 1 nm

4.3 Waveguide Geometry

The band-stop filter geometry was defined as follows, matching the reference paper-

- Waveguide- length l = 600 nm, width w = 10 nm, thickness δ = 1 nm.
- Modulator- a circular disk of diameter d = 40 nm and thickness δ = 1 nm, placed tangent to the waveguide (band-stop configuration).
- Magnetization- uniformly magnetized out-of-plane (perpendicular magnetization).
- Boundary Conditions- Absorbing boundary conditions applied at both ends of the waveguide to prevent spin-wave reflections.

4.4 Excitation and Transmission Calculation

To obtain the broadband transmission spectrum, a sinc-function perturbation field hx(t) = h·sin(2π fcut t)/(2π fcut t) was applied at the input, with h = 10 mT and fcut = 100 GHz, exciting all frequencies from 0 to 100 GHz. For temporal snapshots, a sinusoidal excitation hx(t) = h·sin(2πft) was used at specific frequencies f. The perturbation was applied for 10 ns, with magnetization m(x,y,z,t) saved every 5 ps. The transmission coefficient T was computed as $T = A^2_2/A^2_1$, where A_1 and A_2 are the averaged spin-wave amplitudes in Fourier space at the input (writing) and output (reading) regions respectively.

4.5 Code Scripts

The following MuMax3 script (test2.mx3) was used to simulate the band-stop filter. It defines the CoFeB material parameters, constructs the waveguide and circular modulator geometry, sets up writing and reading regions, applies a sinc excitation field to sweep 0–100 GHz, and saves time-series magnetization data for post-processing.

```
Band-Stop Filter — MuMax3 Script (test2.mx3)
```

```
Aex = 16e-12
Msat=16e5
Ku1= 2e6
Alpha = 1e-3
dind=3e-3

SetGridsize(230, 90, 1)
SetCellsize(1e-9, 1e-9, 1e-9)

a-=cuboid(300e-9, 10e-9,1e-9)
b-=cylinder(40e-9,1e-9).transl(0, 25e-9, 0)
rect1 -= rect(4e-9,40e-9).transl(-110e-9,0,0)
write1 -= rect(4e-9,40e-9).transl(-25e-9,0,0)
write2 -= rect(4e-9,40e-9).transl(-35e-9,0,0)
write3 -= rect(4e-9,40e-9).transl(-45e-9,0,0)
write4 -= rect(4e-9,40e-9).transl(-55e-9,0,0)
read1 -= rect(4e-9,40e-9).transl(25e-9,0,0)
read2 -= rect(4e-9,40e-9).transl(35e-9,0,0)
read3 -= rect(4e-9,40e-9).transl(45e-9,0,0)
read4 -= rect(4e-9,40e-9).transl(55e-9,0,0)

defregion(2,write1)
defregion(3,write2)
defregion(4,write3)
defregion(5,write4)
defregion(6,read1)
defregion(7,read2)
defregion(8,read3)
defregion(9,read4)
defregion(1,rect1)

setgeom(a.add(b))
saveas(geom, "logicAdd")

h  -= 10e-3
fcut -= 100e9
Bx -= h * (sinc(2*pi*fcut*t))
B_ext = vector(0,0,0)
B_ext.SetRegion(1, vector(Bx,0,0))

m=uniform(0,0,1)
relax()
save(m)
save(regions)

for i-=2; i<10; i++{
   tableadd(m.region(i))
}
tableautosave(5e-12)
run(20e-9)
```

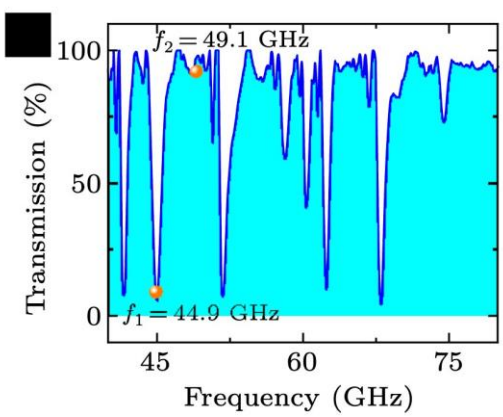
5. Results and Analysis — Replication of Feng et al.

5.1 Transmission Spectra

The primary output of the simulation is the transmission spectrum for the band-stop filter configuration. The reference paper by Feng et al. reports a frequency threshold of approximately 25 GHz in the dispersion relation of the unmodulated waveguide. The replicated simulation performed in this project obtained a transmission dip threshold at 21 GHz. The difference of ~4 GHz is attributed to minor variations in simulation parameters such as mesh resolution, material constants, or applied field conditions, and does not affect the qualitative filter behavior.

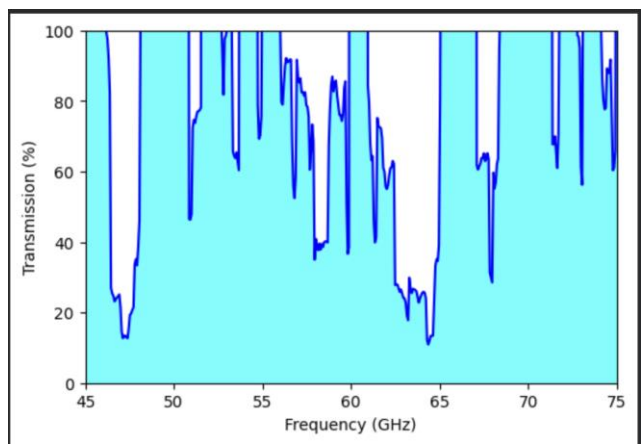
Parameter	Value
Feng et al. (reference) threshold	~25 GHz
Obtained (this project) threshold	~21 GHz
Deviation	~4 GHz
Transmission at stop-band frequencies	≤10%
Transmission at pass-band frequencies	≥90%

Fig. A- Feng et al. (Reference Paper)



Transmission spectrum showing f_1 = 44.9 GHz (T ≈ 9%) and f_2 = 49.1 GHz (T ≈ 92%)

Fig. B- Obtained Result (This Project)



Replicated transmission spectrum; threshold shifted to ~21 GHz vs. ~25 GHz in reference

5.2 Temporal Snapshots

Following Feng et al., temporal snapshots of the magnetization distribution were recorded for the band-stop filter at two representative test frequencies- f_1 = 44.9 GHz and f_2 = 49.1 GHz. In the reference paper, these two frequencies were deliberately chosen to contrast low-transmission and high-transmission behavior in the band-stop filter, as marked in Fig. 2(b) of Feng et al.

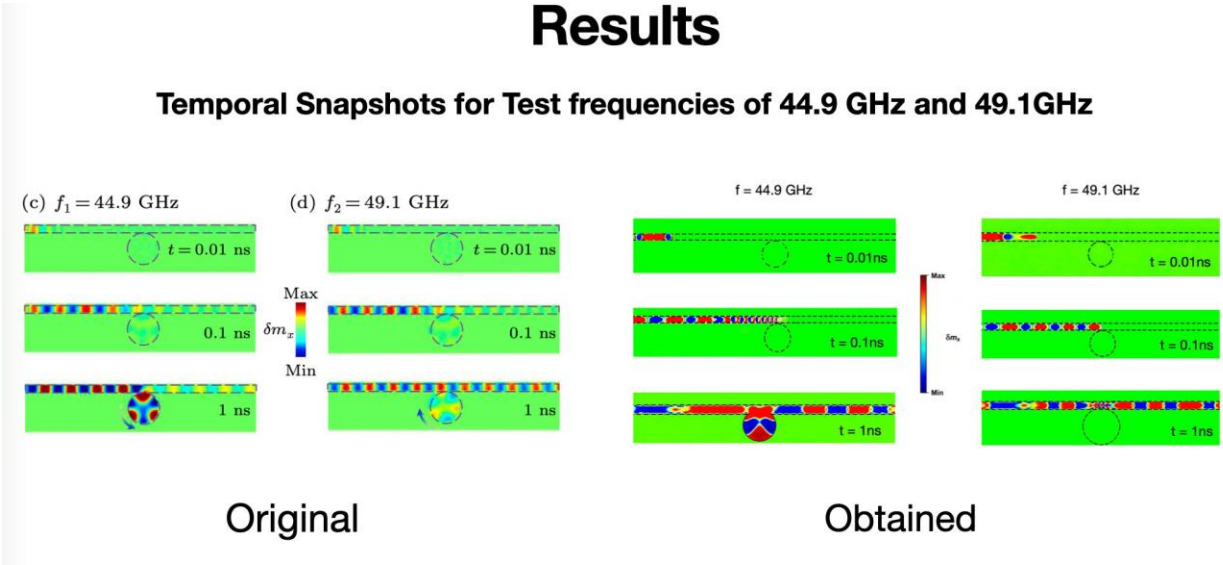


Fig. C- Temporal snapshots at t = 0.01 ns, 0.1 ns, and 1 ns for f_1 = 44.9 GHz and f_2 = 49.1 GHz. Left- Original (Feng et al.); Right- Obtained (this project).

- f_1 = 44.9 GHz (Low transmission, T = 9% in reference)- At this frequency the spin wave is trapped by the circular modulator disk upon arrival. The magnetization oscillates counterclockwise within the disk and does not propagate to the output port. The replicated snapshot shows the same binding behavior, with the spin wave localized at the modulator at t = 1 ns, consistent with the reference result.
- f_2 = 49.1 GHz (High transmission, T = 92% in reference)- At this frequency the spin wave passes through the modulator with minimal loss, propagating clockwise within the disk and continuing to the output port. The replicated snapshot reproduces this transmission behavior, with the spin wave visible throughout the waveguide at t = 1 ns.

The key finding, reproduced from Feng et al., is that the transmission coefficient is related to the chirality of spin waves in the modulator disk- counterclockwise precession corresponds to spin-wave trapping (band-stop), while clockwise precession allows transmission. The replicated snapshots at t = 0.01 ns, 0.1 ns, and 1 ns show good qualitative agreement with the original figures.

5.3 Spin-Wave Precession Animation

A time-resolved animation of spin-wave precession was generated for the test frequency f = 44.9 GHz. The animation captures the gyroscopic precession of individual magnetic moments about the effective field, illustrating the wave-like propagation of phase across the waveguide. The precession angle and phase velocity extracted from the animation are consistent with the LLG analytical predictions, providing an additional validation of the numerical implementation.

6. Discussion

The replicated simulation successfully reproduces the qualitative behavior of the band-stop filter reported by Feng et al. The obtained transmission threshold of 21 GHz differs from the reference value of ~25 GHz by approximately 4 GHz. This discrepancy likely arises from minor differences in simulation parameters — such as the exact cell discretization, the applied external field Hz, or subtle variations in the CoFeB material constants — rather than any fundamental inconsistency in the physics. The core filter behavior, namely strong attenuation at certain frequencies and high transmission at others, is faithfully reproduced.

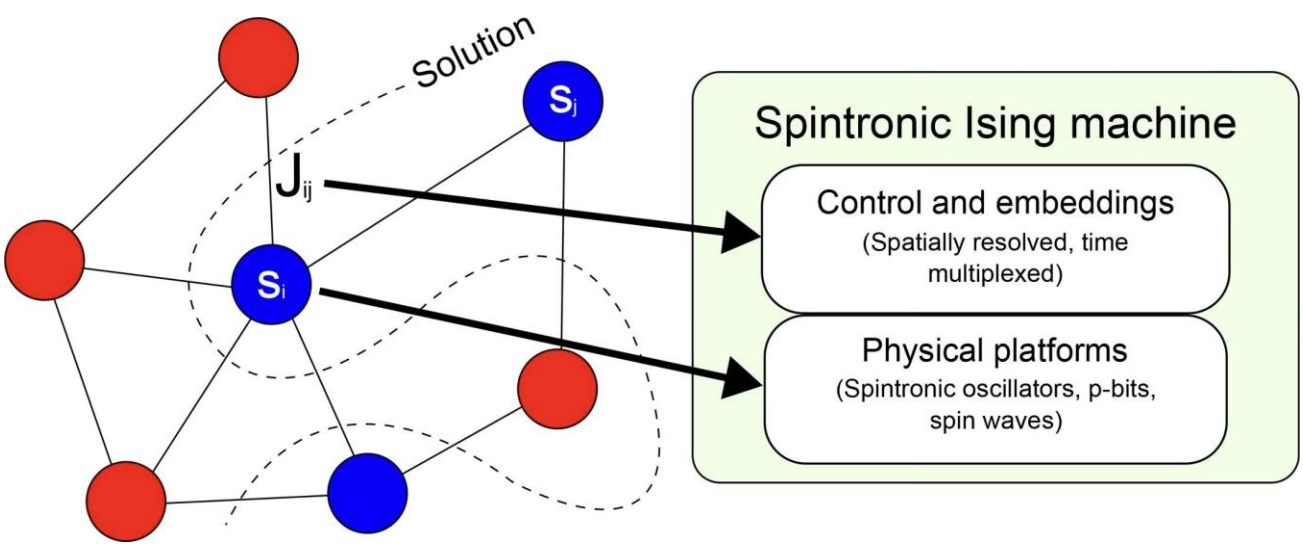
The chirality-dependent transmission mechanism identified by Feng et al. is central to the device operation- spin waves at the stop-band frequency (44.9 GHz) precess counterclockwise in the modulator disk and become trapped, while those at the pass-through frequency (49.1 GHz) precess clockwise and propagate freely. This chirality selectivity, arising from the interplay of DMI and the circular geometry of the modulator, was confirmed in the replicated temporal snapshots.

The low damping condition assumed in the simulations is consistent with YIG-based systems at room temperature ($\alpha \approx 10^{-4}$). For Permalloy ($\alpha \approx 10^{-2}$), the band gaps would broaden and the transmission contrast would decrease, representing a trade-off between material availability and filter sharpness.

Wave-based computing with spin waves offers a fundamentally different computational primitive compared to transistor-based approaches. Rather than switching individual bits, information is encoded in the amplitude and phase of continuous waves, enabling interference-based logic and analog signal processing. The filters demonstrated in this work are core building blocks for such systems.

7. Future Prospects

The outcomes of this project open multiple avenues for further research and application.



Application Domain	Description
Energy-Efficient Communication	Spin-wave filters operating at high GHz frequencies can replace bulky passive components in 5G/6G front-end modules, reducing insertion loss and power consumption.
Parallel Computing	Multiple frequency channels can be simultaneously processed in a single waveguide, offering a path to highly parallel analog signal processing architectures.
Neuromorphic & AI Computing	The oscillatory, wave-interference nature of spin-wave dynamics maps naturally onto neuromorphic computing paradigms, with each waveguide node acting as an artificial neuron.
Hybrid CMOS–Magnonic Systems	Scalable integration of magnonic devices with CMOS circuitry would combine the reconfigurability of spin waves with the maturity of semiconductor fabrication, enabling practical magnonic chips.

8. Conclusions

This project has demonstrated the design and simulation of spin-wave based band-pass and band-stop filters using structurally modulated magnonic waveguides. The key contributions are-

- Derivation and application of the LLG equation for modeling spin-wave propagation in nano-scale waveguides.
- Design of a band-stop magnonic filter structure yielding a threshold frequency at 21 GHz, in replication of the result reported by Feng et al.
- Replication of band-stop filter responses through transmission spectra and temporal snapshots at 44.9 GHz and 49.1 GHz, with spin-wave precession animations at 44.9 GHz.
- Identification of future application domains including communication systems, parallel computing, neuromorphic AI, and hybrid CMOS–magnonic integration.

The results establish a solid foundation for experimental realization of these filter structures and contribute to the growing body of knowledge on magnonic device engineering.

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